

The Emacs Series

Using the f.el API to Work with Files and Directories

In this new GNU Emacs series, we will explore the various Emacs Lisp packages available and also discuss advanced use cases of GNU Emacs.



GNU Emacs is a popular, free/libre and open source software text editor, and is available on multiple platforms. It was written by Richard Stallman in 1985 and the latest stable version is 26.1, which was released in 2018. The source code is primarily written in C and Emacs Lisp, and is released under the GPLv3 licence. The editor is quite extensible and has a large number of packages available for its end users. The project home page is available at <https://www.gnu.org/software/emacs/>. We have already completed an introductory series on GNU Emacs in OSFY and the articles can be obtained from <https://opensourceforu.com/?s=emacs>.

An introduction to f.el

f.el is an API that can be used in GNU Emacs to work with files and directories. It has been written by Johan Andersson and released under the GPLv3 licence. The source code is available at <https://github.com/rejeep/f.el>. The latest tagged release is v0.20.0. It internally requires the use of the s.el string manipulation and dash.el Emacs list libraries.

Installation

The f.el package is available in Milkypostman's Emacs Lisp Package Archive (MELPA) and you can install it using the following command inside GNU Emacs:

```
M-x package-install f
```

You can also copy the f.el source file to your Emacs load path to use it. If you are using Cask, then you can add the following command to your Cask file:

```
(depends-on "f")
```

Usage

Let us go through a few examples to demonstrate the APIs

available with the f.el library.

Path: The entire path to a file can be created using the *f-join* construct. For example:

```
(f-join "/" "home" "guest" "downloads")
"/home/guest/downloads"
```

You can use *f-split* to obtain the individual path elements as shown below:

```
(f-split "/home/guest/downloads")
("/" "home" "guest" "downloads")
```

The entire path of a directory present in the current folder can be fetched using *f-expand*:

```
(f-expand "tmp")
"/home/guest/tmp"
```

You can obtain the file name with its extension using the *f-filename* API. If you need to obtain the parent directory for the file, you can use the '*f-dirname*' construct. The file name extension can be fetched using the *f-ext* API as illustrated below:

```
(f-filename "/etc/resolv.conf")
"resolv.conf"

(f-dirname "/etc/resolv.conf")
"/etc"

(f-ext "/etc/resolv.conf")
".conf"
```

If you would like to rename a file name extension, you